BrainAge Model Architecture

14 per-channel contrastive encoders → cross-channel attention → age regression

A. EEGConvGRUEncoder (per channel, pretrained)

EEG channel input

[B, 512, 1]

Conv1d → BatchNorm → Activation → Dropout

1 → C channels, kernel k, stride s → [B, C, L′]

Permute → GRU / LSTM

[B, L′, C] → hidden size H, num_layers → last h_n [B, H]

Projection head: Linear(H → D) + L2 Normalise

contrastive embedding [B, D]

Channel Projector: Linear(D → 16) + LayerNorm

[B, 16] — per-channel feature fed into AgeModel

B. AgeModel (full pipeline)

EEG Input

[B × 512 × 14] (14 channels, 512 time steps)

EEGConvGRUEncoder × 14 (pretrained)

one independent encoder per channel → embedding [B, D_ch]

Channel Projectors × 14

Linear(D_ch → 16) + LayerNorm → [B, 16] per channel

Stack channel features

[B, 14, 16]

Multi-Head Self-Attention (across channels)

embed_dim = 16, heads = 2 → [B, 14, 16] + weights [B, 2, 14, 14]

Flatten

[B, 14 × 16] = [B, 224]

FC1: Linear(224 → 512) + Dropout(0.1) + SiLU

FC2: Linear(512 → 512) + SiLU

Output: Linear(512 → 1)

Predicted Brain Age (years)

[B] — regression output